was also seen as the mascot of this propounded Indian space philosophy. But probably more Americans, than Indians, knew that Sarabhai's dreams extended to making India a missile power.

A thin, not-so-hazy, line separates satellite launch vehicles from missiles, as both are essentially rockets. One has a satellite as the payload, the other has a warhead. One is designed to go into space and never come back, the other may go into space or to a lower altitude before landing back on earth, at a predetermined target. One develops, the other destroys.

I sensed Sarabhai's interest in missiles as early as my first interactions with him. If pursuing liquid propulsion system was my passion, persuading our people, including Sarabhai, to believe in it wasn't easy. There were times when I wondered why Sarabhai, such a visionary, glossed over my arguments for developing a liquid engine, while he encouraged solid propulsion projects. Sarabhai had bought the PVC propellant making technology from France and set up the Rocket Propellant Plant (RPP) in Trivandrum. M R Kurup, who headed RPP, became a natural 'ally' of Kalam who was working on the solid-propelled SLV, in the solid-liquid tug-of-war. I was alone.

I found reason in the argument that we cannot ignore solid, since the technology was available and rockets were indeed fired using the propellant, but what troubled me was the indifference to liquid propulsion systems, which were far superior though more difficult to develop. It took a while for me to realise that Sarabhai's support for the solid group had another core reason: solid propellants, he believed, were inevitable for developing missiles.

I heard this from the horse's mouth once when I was arguing my case in private with Sarabhai.

'But missiles are made with solid propellants,' he remarked.

I got the message. I spent longer hours in the library to find a convincing reason to push for my pet project. I went back to Sarabhai and impressed upon him that liquid not only did work for missiles, it was imperative if India was to develop intercontinental ballistic missiles that would go to space before returning to earth to strike targets anywhere on the globe.

The US wasn't ignorant of Sarabhai's network of friends. Even before trying for an ISRO-CENS France tie-up, Sarabhai had forged an ISRO-DFVLR programme with Germany. This brought German scientists like